Visible flaws

A comprehensive quality assessment of Visible Learning

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Abstract

John Hattie's *Visible Learning* aggregates over 2,400 meta-analyses into ranked effect sizes, claiming to reveal the most impactful educational practices. Despite its global influence on education, critics highlight methodological and statistical weaknesses. This study comprehensively assessed *Visible Learning's* 322 influences for quality, relevance, and reproducibility. Findings reveal that only 19% meet the inclusion criteria regarding participants, intervention, comparison, and outcome alignment (PICO). Furthermore, in the attempts to reproduce the extractions, only 2 out of 301 influences were successful. The assessment revealed that *Visible Learning* contained methodological errors that compromised both the causal interpretations and the validity of the reported effect sizes. An expert panel found that while 23% influences are highly relevant, many hold little relevance to K-12 education. These findings underscore significant methodological and statistical shortcomings, raising questions about the validity of *Visible Learning* as a basis for educational policy and practice.

Keywords: Visible Learning, Educational research, Meta-analysis quality, Reproducibility, Methodological assessment.

Author contributions

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The main part of the present research was conducted by the core project team, lead by first-author A.K, and with T.N and R.C as the senior authors. J.D, J.H and D.E served on the expert panel that conducted the relevance checks. They also provided substantial comments and edits to a first draft of this article. Below, we further detail the contributions with CRediT.

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Introduction

In 2009, John Hattie released the meta-meta-analysis *Visible Learning* which summarized 800 meta-analyses into 138 possible influences on student achievement (Hattie, 2009). The influences were all re-coded to a standard metric (Cohen's *d*) and ranked based on their effect sizes, ranging from negative (e.g. retention), little effect (e.g., student personality), to strong influences on student achievement (e.g., Response to intervention). The *Visible Learning* dataset grew quickly and by 2021 it contained 322 influences based on over 2400 meta-analyses (Hattie, 2023). It further contained guidance for readers without statistical expertise on how to interpret the effect sizes.

The availability of an "easy-to-comprehend" overview of the research that revealed both positive and negative effects in school was welcomed with gratitude by many professionals working in schools. The ranking is now used in over 23 countries around the world and the original book received over 22,000 citations on Google Scholar which means that the impact of the work cannot be overstated (Knudsen, 2017). It has also gained scholarly attention well beyond the anglosphere tradition from which it came (e.g., a special issue in the Nordic scholarly journal *Paideia* (Hansen et al., 2015). It is also safe to say that Hattie's work has helped accelerate disparate educational research studies into an evidence-based compendium of quantitative analysis.

Despite this*, Visible Learning* was met with a lot of skepticism and was considered controversial from the start. The basic method of *Visible Learning* is to aggregate all available meta-analyses on a great number of candidate influences on student achievement (e.g., classroom size) into one effect size per influence, and then compare these aggregated effect sizes to each other. According to Hattie (Nejman, 2023), the strength of his work is the ability to combine all kinds of studies through time and context, but several researchers disagree with their methodological approach (Arnold, 2011; Knudsen, 2017; McKnight & Withburn, 2020; Nielsen & Klitmøller, 2021; Simpson, 2017; Slavin, 2018; Snook et al., 2009). The criticism was fierce, and some even qualify the work as pseudoscience (Bergeron, 2017), due to the lack of conventional scientific methods and principled statistical analyses.

Although the criticism against the methodology in *Visible Learning* has been severe, there have been no systematic and exhaustive attempts to formally assess it. Rather, the critique has focused on pointing out specific errors and problems. Of course, a gigantic project like *Visible Learning* cannot be error free, or to solve all complex conceptual issues. At this point, it is not possible to tell if this criticism is based on cherry-picked problems from an overall solid research effort (DeWitt, 2018), or just the tip of the iceberg.

Our own view is that *Visible Learning* has great potential if it is correctly executed. We also think that any scientific endeavor with this level of impact deserves to be taken seriously and be properly evaluated. To this end, the overall purpose of the present study is to conduct an exhaustive and systematic assessment of the literature review methods and meta-analytical results of *Visible Learning*. The result of our comprehensive assessment can lead to different conclusions on the best way forward. The issues previously identified might be limited to a few

influences and easily fixed. Conversely, the issues might be systematic and require a major, or even complete, overhaul of *Visible Learning* before it can be relied upon by policymakers and professionals working in schools.

In the remainder of the introduction we will review the previous efforts to assess *Visible Learning.* We will also introduce the current best practices for how to systematically assess this type of meta-review. The specific aims of the present research concludes the introduction.

Previous criticism against Visible Learning

Since its introduction, several scholars have pointed out problems with *Visible Learning*. According to Nielsen and Klintmøller (2021), *"...the theoretical ideas of visible learning are narrow* (i.e., reducing a complex activity like teaching to a single effect size measure) *and risking that all dialogues about learning, education, and teaching become obsolete"*. In particular, the problem is that the effect sizes presented are taken out of their contexts, and thus become impossible to interpret. Nielsen and Klintmøller (2021) highlight the concept class size as an example and refer to Bergers' (1982) statement that class size should be understood in relation to several variables such as students' ability, social background, teacher's ability to teach, and teacher's resources and cannot be understood as a meaningful analytical unit in itself. Nielsen and Klintmøller (2021) investigated meta-analyses connected with Hattie's coverage of the influence *'feedback'* and discovered that it was unclear whether they used a feedback term identical to Hattie's description in *Visible Learning* - two out of the five meta-analyses did not even mention feedback.

Simpson (2017) argues that using ranked meta-meta-analyses to drive educational policy is misguided. The rationale of this statement is that the numerical summaries that are used to develop these results are not a measure of educational impact but instead a result of research areas that are able to design sensitive experiments. With this said, Simpson (2017) states that quantitative summaries are a good representation of intervention effectiveness but their rank order does not represent better and worse examples of 'what works', in other words, "*... a larger effect size does not indicate a larger educational impact for the intervention*".

The argument by Simspon (2017) is necessarily true since standardized effect sizes have the study's variance (or standard deviation) as the denominator. For example, Cohen's *d* is the raw mean difference between two conditions divided by the standard deviation. It matters a lot if, for example, that standard deviation is the standard deviation across individuals or within individuals. The between participant standard deviation is often considerably larger than the within-participant standard deviation. This means that effect sizes from widely different designs cannot be combined or compared without careful attention to how they were calculated, and appropriate adjustments made to bring them to a comparable metric. Importantly, *Visible Learning* is not taking any such steps in their methodology.

However, the meaning of an effect size does not only depend on the design, but also other contexts, such as the populations. Consider for example evaluations of reading interventions, such as the model Response to intervention (Nilvius et al., 2021). The model targets students

with difficulties learning to read, delivering support in smaller group settings and comparing results to the standard classroom or school practices. Effect size, often measured by Cohen's *d*, can sometimes be statistically small (around ~0.3), even when interventions are deemed effective. This modest effect size is likely due to the nature of the challenges these students face. Struggling readers may have persistent and significant difficulties, and interventions, while helpful, may not result in drastic improvements. Small effect sizes in this context do not necessarily indicate failure but rather reflect the complex nature of reading difficulties and the ceiling of potential short-term gains that can be expected from targeted interventions.

Arnold (2011) questions the validity of the approach of *Visible Learning* and argues that the relationship between the narrative in *Visible Learning* and the effect sizes is not transparent and clear. After having looked deeper into the included meta-analyses, Arnold discovered that one of the top ranked influence (self reported grades) was dominated (highest effect size) by a meta-analysis that did not even closely cohere to the definition of the influence, and such, raised concerns regarding the average effect size presented in all of the other influences.

Snook et al. (2009) comments on the quality aspects of the studies included in *Visible Learning* and state that any meta-analysis that does not exclude poor or inadequate studies is misleading and might lead to ill-advised policy developments. Another critique put forward is the difficulty that arises when averaging a large number of disparate studies, important variables such as age, ability, gender, and even subjects studied are set aside. Snook et al., (2009) takes the influence '*homework'* as an example; The overall effect size is 0.29, however when looked up in detail the effect size differs between age groups (0.15 for elementary school and 0.64 in high schools), between subjects (mathematics showed high effects and science and social studies showed low effects sizes), high ability students showed low effects and low ability students showed high effects. Clearly, the potential influence homework is a complex matter and all these variations are lost in an average effect of 0.29, which is below the 'hinge point' of 0.4 that *Visible Learning* has used a cut-off of when an influence is meaningfully large (Hattie, 2009; 2023).

To conclude, some researchers have found examples of meta-analyses included in influences that make the results invalid and question the validity of this work. Others question the methodological underpinnings of *Visible Learning*. Hattie describes his role in *Visible Learning* as the one who updates the meta-analyses and discovers the errors (Knudsen, 2017) but the examples given above are still included in the analyses years after discovery. Anyone can understand that when combining and extracting over 2000 meta-analyses the work cannot be entirely free from errors. However, without an assessment of how common such errors are, we believe that this leaves school professionals and policymakers with an unacceptable uncertainty of the quality of this work.

Best practices in assessing meta-reviews

A meta-review (or meta-meta-analysis) is not conceptually different from a meta-analysis or a systematic review. Instead of calculating the average standardized mean difference (usually

Cohen's *d* or Hedges' *g* in social sciences) among a set of original studies, a meta-meta-analysis calculates the average from a set of meta-analyses. It is well known that meta-analysis cannot improve much beyond the studies they include: the garbage in garbage out principle. Furthermore, studies need to be identified, evaluated and included in a systematic manner to avoid bias (e.g., cherry picking favorite studies). To this end, high-quality meta-analyses are conducted as the final step of a systematic review. A systematic review uses a clear set of rules to determine which studies will be included or excluded from the analysis. This provides a transparent, objective, and reproducible framework for the discussion on the substantive meaning of the results (Higgins et. al., 2024). Similar to how individual meta-analyses need to follow systematic review methodology, so does a meta-meta-analysis.

In order to fully grasp the quality of *Visible Learning* we need to assess this work to the current standards of conducting a systematic review with meta-analysis. Carlsson et al. (2024) explains three standards when conducting a systematic review (with meta-analysis) and recommends the Methodological Expectations of Campbell Collaboration Intervention Reviews (MECCIR) for social sciences, including education science.

The MECCIR guide (Dewidar et al., 2023) states that in order to conduct a systematic review, the eligibility criterias are a fundamental prerequisite for a systematic review. The eligibility criterias are the deciding factors that drive what publications should be included in the analysis and are tied to the targeted population, intervention, comparator, outcome, and study type (PICOS). Below follows a short rationale of the eligibility criterias as well as the importance of reproducibility and relevance.

Population. Capturing the right population is an important step in any review. For example, the results from a reading intervention conducted at the university level are not easily transferred to students in elementary school and would therefore not be included in a typical meta-analysis. If we want to generalize the findings of a meta-analysis we first need to define the relevant targeted population; if the population is not representative of the studies included in the synthesis the estimates will be biased (McShane et al., 2016). According to MECCIR (Campbell Collaboration, 2014) "*the criteria for considering types of people included in studies in a review should be sufficiently broad to encompass the likely diversity of studies, but sufficiently narrow to ensure that a meaningful answer can be obtained when studies are considered in aggregate"*. We want to understand if *Visible Learning* has a meaningful answer to effect sizes regarding students in grades K-12 and therefore a population outside this eligibility criteria (eg. adults or university students) are not as relevant.

Interventions. The selection of interventions grouped for synthesis is most likely to influence the review findings and by clearly defining the sought intervention enables a comparison and provides a consistent language for reporting that helps interpretation of the findings (Higgins et al., 2024). For example, the effect of a feedback intervention conducted verbally in groups may differ from the effect of an intervention conducted one to one. Not defining the eligibility criteria of the interventions will lead to more variation in the intervention

effects (increased heterogeneity) and might weaken the accuracy or remove the validity of the results as mentioned in a previous section by Arnold (2011).

Comparison group. Campbell Collaboration (Aloe et al., 2024) outlines specific eligibility criteria for comparison groups, an important step to follow to make informative aggregations in the review. Let us consider the effect size of increasing a class size by 10 students with one study comparing 2 students to 12, and another 20 students to 30. Do these two effect sizes capture the same phenomena (i.e., are these two differences qualitatively similar)? Does it make sense to aggregate these two into one mean effect size? MECCIR (Campbell Collaboration, 2014) requires mandatory specification of comparator interventions, including the extent to which the experimental interventions are compared with a control or comparison condition with matched or similar participants. For example, you should typically not aggregate a comparison between two experimental conditions with a comparison between an experimental group and a control group. But, unequal comparison groups are commonplace in the studies included in educational meta-analyses and meta-meta-analyses (Allison et al., 2016; Simpson, 2017). The present study seeks to map out if and to what extent unequal comparison groups are aggregated in *Visible Learning*.

Outcomes. When conducting a systematic review it is important to specify the outcome measures. Predefining outcomes allows users to confirm that choices were not overly influenced by the results (Higgins et al., 2024). It is common for studies to also include different aspects of the intervention such as adherence, motivation or how much time is spent on different intervention stages. The total number of outcomes selected for inclusion in the review should be kept as small as possible since large numbers of outcomes can make reviews unfocused and prone to selective outcome reporting bias (Campbell Collaboration, 2014). For example, if we are interested in student achievement and combine test scores with these auxiliary measures mentioned above we risk making the review unfocused and hard to interpret.

Study type. Study type is a quite large part of the PICOS mnemonic, because studies can differ in many ways. For example, if they are qualitative or quantitative, what discipline they are conducted in, and if they are empirical or theoretical work. In *Visible Learning*, many parts of study type are already given, as a meta-review only includes reviews of empirical research. We have thus narrowed this part down to a few sub categories that we found most relevant when assessing *Visible Learning*: study design, publication type, and year of publication.

Design: we can divide the study designs into two categories: experimental and observational. (1) experimental studies (randomized controlled studies (RCT) or quasi-experimental studies (QES)) are used in order to test a causal relation between our variables. The RCT is the idealized study design to use for evidence based practices (Campbell Collaboration, 2014), but sometimes randomization of participants cannot be done for practical purposes, or randomization might increase other problems in the study (e.g., uneven attrition) to the point that the internal validity is threatened. In these cases, QES is the best option. We know that quasi-experiments can sometimes overestimate effect sizes compared to randomized experiments in educational research because they cannot fully control for confounders the same

way an ideal RCT can (Cheung & Slavin, 2016). Hence we have to be aware of this risk of overestimation when interpreting the results. This risk will vary greatly depending on the quality of the studies and on the contexts, and it is important to emphasize that although RCT is in theory a stronger design than QES, there are many other aspects of studies that affect their quality. Hence, whether it makes sense to combine QES and RCTs into an overall aggregate, or analyze them separately, will be different for different topics, but it is something that should be specified in advance.

Observational studies are used to understand a phenomenon that can not be designed in an experiment. For example we know that bullying can affect school performance but we can not actively bully a group of students in order to measure the causal effect. This has to be studied in other ways, however, observational studies can not provide proof of a causal relationship between the variables and the results should not be treated as such. Observational studies and experimental studies should never be simply averaged together into a meta-analytical aggregate. If they must be combined it needs to be through a principled framework for combining different types of evidence.

Different types of designs yield different types of effect sizes and the most common measures in educational research are correlational and standardized mean differences (SMD). According to Borenstein et al. (2009) converting effect sizes are necessary when studying broad questions and want to include several types of effect size measures into one analysis. However, the question regarding whether it is appropriate to combine these effect sizes must be considered on a case by case basis. For example, observational studies that report correlations may be substantially different from randomized control studies using standardized mean differences and combining these effect sizes is almost always a bad idea from a substantive perspective (Borenstein et al., 2009), and this is also not a common practice. It is less commonly known by meta-researchers that different experimental designs often estimate different population parameters and cannot, unless adjusted, be compared or aggregated (Morris & DeShon, 2002). Today's standards do not provide a simple solution regarding how to deal with independent-group and repeated measures designs in the analysis, and it will thus depend on the context whether it is best to provide separate estimates for different designs or adjust the effect sizes to make them more comparable. However, there is a consensus that they cannot be naively combined without addressing the differences in some way.

In educational research, we need all types of study designs for different reasons, but we believe that one should not mix study designs (eg. within- and between subject-designs, or experimental and correlational designs) and expect that they are providing the same type of results just because they provide the same measurement unit (effect size). Importantly, *Visible Learning* is combining effect sizes through simple unweighted mean averages, which can never be an appropriate way to combine anything but highly comparable effect sizes.

Publication type: In order to capture a phenomenon we need to collect all available information, from both the published and non published literature, otherwise we risk publication bias (McShane et al., 2016). Research has shown that only including published studies may

inflate the effect size (McShane et al., 2016; Simonsohn et al., 2014). Conversely, if we are heavily relying on non-published literature (such as dissertations or conference papers) we miss out on the quality aspects of peer-review. All guides in meta-analytic research are meticulous regarding the comprehensiveness of the searchers in order to make sure that all research is captured in each given field.

Publication year: The year of publication may also play a role in certain research questions. For example, studies examining the effect of computers in school made in the 1980 might not be as representative as studies conducted today. Another example is that several meta-analytic studies may include the same original study in their synthesis, this leads to unequal weight of the studies since some results are counted twice or more. By collecting descriptive information regarding the publication type we can create a holistic picture regarding the representativeness of the included studies in *Visible Learning*.

Reproducibility. Meta-analyses are often regarded as the state-of-the-art method to provide conclusions regarding specific effects or research areas and it is therefore crucial that they provide the highest possible quality (Lakens et al., 2016). Some argue that quality can be ensured with the peer-review process, however, studies showed that effect sizes derived from meta-analyses can be hard to reproduce (Gøtzsche et al., 2007), which lead to low credibility of conclusions (Derry et al., 2006). We thus want to investigate if the effect sizes in *Visible Learning* are reproducible in order to ensure that the results are trustworthy.

Relevance. Well-conducted and reproducible meta-analyses will not be of much value if they are not on relevant topics. *Visible Learning* is quite exhaustive in what types of influences are included in the list. This great heterogeneity makes comparisons even harder. It spans from influences such as breast-feeding, chess instruction and phonics. We do not believe that it is useful for teachers and policy-makers to be overwhelmed by a large number of peripheral influences. One way forward towards finding influences that can, to some extent, be compared (although not numerically) is to focus on important and relevant influences. This will also present us with a "short-list" of areas to focus addressing errors (if any) in.

The present study

To our knowledge, there has been no systematic attempts to investigate the quality of the methods or reproducibility of the meta-analytical statistics extracted of all included influences presented in Hattie's *Visible Learning*. To this end, the purpose of the present study is to conduct an exhaustive and systematic assessment of *Visible Learning*. We will conceptually and critically review the methodology, examine the quality of the coding, and have an expert panel of educational researchers assess the relevance of the included 322 influences in *Visible Learning*.

We want to understand what type of population, interventions and outcomes that are included in *Visible Learning* in order to know if this work is relevant for children in school (K-12), which is the most common population that are affected by *Visible Learning*. Based on previous statistical and methodological criticism we also want to see if we can reproduce the statistical findings in *Visible Learning*.

In order to assess the relevance, quality of the aggregation and coding of *Visible Learning* seven quality indicators will be used:

1. To what extent are the Participant, Intervention, Comparison, and Outcomes (PICO) in the included meta-analyses relevant to the definition of the influence?
2. To what extent have randomized controlled trials (RCT), quasi-experimental studies (QES), and observational studies been mixed?
3. To what extent have effect sizes from different designs (within-, between-subject designs, or correlational research) been mixed?
4. To what extent have meta-analyses with a wide publication year range been mixed?
5. What publication types (conference papers, unpublished doctoral theses, peer-reviewed articles) are included in *Visible Learning?*
6. To what extent can the reported statistics in *Visible Learning* (effect sizes, numbers of studies, or participants) be reproduced?
7. What is the perceived relevance of the different influences in *Visible Learning*, based on a panel of expert education researchers' ratings?

It should be noted that we limit our assessment to relevance, literature review methodology and reproducibility of meta-analytical results, and that we will not assess other aspects of *Visible Learning* such as theoretical, epistemological or practical, but see (McKnight & Whitburn, 2020; Nielsen & Klitmøller, 2021; Rømer, 2019; Wrigley, 2018) for such discussion. Importantly, our aim was to provide a fair assessment of *Visible Learning*, which is why we pre-registered our study to avoid the risk of cherry picking, or moving the goalposts for what criteria should be assessed.

Method

Pre-registered protocols

This study has been pre-registered and the protocol, data and supplementary materials can be found here:
Project OSF: https://osf.io/9w7rv/
Protocol Quality assessment: https://doi.org/10.17605/OSF.IO/3N7GH
Protocol Relevance assessment: https://doi.org/10.17605/OSF.IO/AVSGC

Differences between the protocol and the final report are detailed in the supplemental materials and anything but very minor deviations are included in the paper.

Software

All code was produced using R.4.2.2 (R Core Team, 2020), and the R packages: DataTables (Xie et al., 2023), data.frame (Vaidyanathan et al., 2023), dplyr (Wickham et al., 2023), ggplot2 (Wickham, 2016), janitor (Firke, 2023), patchwork (Pedersen, 2022), pdftools (Ooms, 2023)

psych (William, 2024), shiny (Chang et al., 2023), shinywidgets (Perrier et al., 2024), skimr (Waring et al., 2022), stringr (Wickham, 2022), tidyverse (Whickham et al., 2019).

Data collection process

All data collection was retrieved by the first author and sorted into a Zotero library. The influences in *Visible Learning* (version 1.2) is published online in the global research database (Corwin Visible Learning Plus, 2023). Each influence consists of metadata such as definition, domain, included statistics of numbers of meta-analyses, studies, students, and effects. The included meta-analyses of each influence are described by journal title, author, first author's country, article name and year published. The extracted effect size of each meta-analysis is described by the variable name, numbers of studies, students, and effects.

Searches

In order to identify the meta-analyses included in each influence the first author searched the following databases for article name, author, published year and journal title during 2021/03/17-2023/09/19:

- APA PsycInfo (ProQuest)
- ERIC (ProQuest)
- Linguistics and Language Behavior Abstracts (ProQuest)
- MLA International Bibliography (EBSCO)
- Web of Science: Core collection (Clarivate)
    - SCI-Expanded (1900-present)
    - SSCI - (1955 -present)
    - AHCI - (1975 - present)
    - ESCI - (2005 - present)
- Google Scholar
- Research gate

Retrieval of meta-analyses

A total of 1369 meta-analyses could be retrieved, 620 papers were missing. The majority of the missing meta-analyses were derived from several unpublished dissertations, thus making it impossible to retrieve. A total of 395 duplicates could be found. In cases of duplicates, data extracted from a meta-analysis could be used for different influences and sometimes several times within a single influence extracting multiple effect sizes from different variables or outcomes. Deviation from protocol; a PRISMA flowchart of identified and/or included studies is not provided. The reason is that we did not find it particularly useful as an illustration.

Assessing relevance

A total of nine international researchers, within the field of education, were invited and five researchers committed to conduct the relevance check. We did not pre-register the criteria for inviting panel members, but relied on our knowledge to find experts who would together have the broad competence to be able to assess *Visible Learning*. Further, we aimed for the researchers to be as independent from our own research networks as possible (i.e., no previous co-authorships). One researcher left the panel during the progress due to being unable to meet the deadline. We had pre-registered that we would recruit a new panel member if anyone would leave the panel. However, as the experts had started their work, we held out hoping for them to finish for too long, and it was no longer feasible for us to justify delaying finalizing the study. Three of the panel members agreed to contribute with writing and editing of the final paper and to be listed as co-authors.

The expert panel answered five questions regarding the influence based on the information provided in the *Visible Learning* global research database (Corwin Visible Learning Plus, 2023). These questions refer to if the presented interventions, methods or strategies are *concerning the typical student*, if it is *widely adopted* and if it is *relevant to schools*. All questions used by the expert panel can be found in the supplementary materials. The intraclass correlation coefficient for judges rating of the relevance, ICC2K = 0.60, CI 95% [0.53, 0,67], can be considered moderate to good (Cicchetti, 1994; Koo & Li, 2016).

Assessing quality

The first author read all meta-analyses included in each influence and assessed the quality by answering questions regarding the PICO, study- and participant-design, year of publication and publication type. The two senior authors validated 50% of the influences combined, and any discrepancies between researchers were resolved through discussion.We had pre-registered that 100% of the influences were to be validated; however, validating the quality assessment was more time-consuming than expected. Hence, we decided to stop at 50% and ensured that the validation was equally spread out over the categories. Twelve dichotomous questions with binary answers (Yes/No) were used to assess the quality (Eg. "*Does the meta-analysis include a relevant population for the influence*?" or "*Does the meta-analysis include randomized controlled trials?"*).

We do not present any indices of inter-rater agreements. The reason is that the very large scale of this project made it unfeasible for the three raters to rate the entire dataset independently and then compare afterwards. Instead, we worked independently (and blinded), in parallel, and compared results after we had rated several influences, but not more than that we could remember the content. As such, any index would be biased as we worked towards achieving consensus in the ratings, and thus had less disagreements later on in the process. Further, the extreme heterogeneity of the material rated means that a single index would be quite meaningless, as the difficulty of the task varied from trivial to very difficult due to a great

number of factors: clarity of the influence included, type of influence, number of studies included, clarity of studies included.

All in all, our qualitative impression is that we had high substantial agreement for almost all influence ratings, but that we very often had some minor disagreement (e.g., did one of the included studies really include a quasi-experimental study or not). As we share the full rating sheets, it could be an interesting task for some specialists to model our inter-rating agreements, but that is beyond our capabilities.

Descriptives regarding the publication year and type were also coded by the first author only. We started double coding this, but quickly found it to be redundant as the coding is a simple matter of copy and paste with little or no judgment. All questions can be found in the supplementary materials.

Reproducing the extraction of meta-analyses

In order to reproduce the extracted statistics, each meta-analysis was randomly ordered within each influence by a random number generator. The extracted statistics inherent from the meta-analysis were reproduced until failure, with failure meaning that the researchers could not reproduce either the effect size, number of effects, studies, or participants presented in *Visible Learning*. First author reproduced all influences and two independent researchers (the senior authors) validated 50% of the influences combined. All disagreements were resolved between the first author and the validating researcher. Codesheets regarding reproducing the extraction of meta-analyses can be found in the supplementary materials.

During the phase of extraction we realized that an additional category was needed in the code sheet, hence an update to the protocol was made. We realized that several effect sizes could be found but were derived from statistical calculations that were outside the stated definition of influence, therefore we added a third response alternative to the question *Can the effect size be reproduced? Yes - Does not match PICOS.*

Shiny application

Due to the amount of data included in this study we found that a normal table is hard to decipher and navigate through. Therefore, we created an interactive shiny app with filtering options for all data. Throughout the result section we will show examples of our findings but the results of all influences can be found in the application or the supplementary material..

Results

The result section followed the analysis plan in the pre-registration without any deviations. Importantly, our preregistration made explicit that we do not have any hypotheses, and thus no corresponding statistical tests, and that we would only present descriptive statistics. The individual assessments are available in the online supplement (https://osf.io/9w7rv/) in the form of spreadsheets. To facilitate easy exploration of this vast material, we have also created a Shiny App that is linked to in the online supplement.

This result section has the following outline. First, we review the results in the same order as we presented the questions earlier. Note that this order does not reflect the order of importance of the questions. For each question, we provide the overall result, as well as an illustrative example of the issues found during the assessment. Note that these examples are not criticism of the specific studies included in *Visible Learning*, as we have only assessed *Visible Learnings* synthesis of them and not the studies themselves.

A total of 301 (out of 322) influences were assessed, these influences consisted of 1369 retrieved meta-analyses. We procured full texts, read, and assessed them all. For 21 of the influences we were unable to find any included meta-analyses, and they are therefore treated as missing in this part of the assessment. We were unable to retrieve 620 papers which resulted in a total of 180 influences not being complete but still assessed without the missing papers.

1. To what extent are the Participant, Intervention, Comparison, and Outcomes (PICO) in the included meta-analyses relevant to the definition of the influence?

An overwhelming majority of the influences included irrelevant studies on at least one of the PICO criteria, with only 19% (n = 56) passing our assessment of all criteria. As many as 25% (n = 74) failed our assessment across all the criterias of PICO. Table 1 details the extent that each influence included relevant studies.

Table 1
Inclusion of relevant studies (PICO) in influences

Relevant	Population	Intervention/Exposure	Comparison	Outcome
Yes	160	140	123	124
No	141	161	178	177

Population. It was common for influences to include the wrong population. In total, 53% of the extracted meta-analyses had students in K-12 as the targeted population in their studies. The majority of the influences assessed as not having a relevant population included studies with a sample collected from university students or working adults. An example of non relevant inclusion can be found in the influence '*Depression'* where Huang et al. (2014) was included in the synthesis. This original study focuses on adolescents' maternity depression as a mediator for infant development, which is the wrong population. This study is also not a meta-analysis.

Intervention (or exposure). It was common to include the wrong intervention (or exposure), and our assessment showed that only 46% (n = 140) of influences passed this check. For example, in the influence *'Writing programs' Visible Learning* included a meta-analysis conducted by Graham et al. (2018) which is examining a reading intervention rather than a writing program.

Comparison. 41% (n = 123) of the influences were assessed as having included relevant comparison groups. In the majority of the influences, the results were derived from studies

comparing two or more interventions instead of a control group. The interpretation of an effect size comparing two interventions is different from a comparison to a control group that consists of "teaching as usual". For example, if a highly effective intervention is compared to another highly effective intervention, the resulting effect size will be close to zero, and thus misleading as it is large when compared to a control group. Although this can be very interesting in itself (e.g., equivalence trials), it cannot be pooled together with intervention studies compared to conventional control groups.

Outcome. 41% (n = 124) of the included meta-analyses turned out to have student achievement as outcome. The majority of the influences synthesized meta-analyses with completely different outcomes, or extracted the overall effect of an intervention and not only the achievement score. There are several examples of when motivation, behavior or IQ is included in the extracted effect size. For example, in the influence '*Response to intervention'* the extracted effect sizes from Tran et al (2011) combines behavior, IQ and performance; Stuebing et al., (2015) correlates children's characteristics towards gain with RTI; and Burns and Symington (2002) uses a within subject design with measures such as observed behavior among university students.

2. To what extent have randomized controlled trials (RCT), quasi-experimental studies (QES), and observational studies been mixed?

Four influences (1%) were based upon meta-analyses including RCT studies solely, 9% (n = 28) QES, and 25% (n = 74) were based on observational studies. Further, 36% of the influences (n = 108) combined RCT and QES study designs in the aggregation.

A total of 82 influences (27%) mixes RCT and/or QES with observational studies. In these cases, as mentioned in the introduction, the causal interpretation of the effect sizes is questioned by combining experimental data with observational data due to that observational data is more subject to unobserved confounding.

3. To what extent have effect sizes from different designs (within-, between-subject designs, or correlational research) been mixed?

A total of 89 influences (30%) aggregated meta-analyses of between-subject designs solely, two influences (<1%) of within-subject designs, and 53 influences (18%) were based on correlational research.

The assessment revealed that 10% of the influences aggregated effect sizes derived from all types of between-, within-subject designs and correlational research. 17% mixed between- and within-subject designs and 21% mixed either between- or within-subject design with correlational research. Further, 9% of the influences did not provide enough information to assess where the effect sizes derived from. The extent to which effect sizes have been mixed can be found in Table 2.

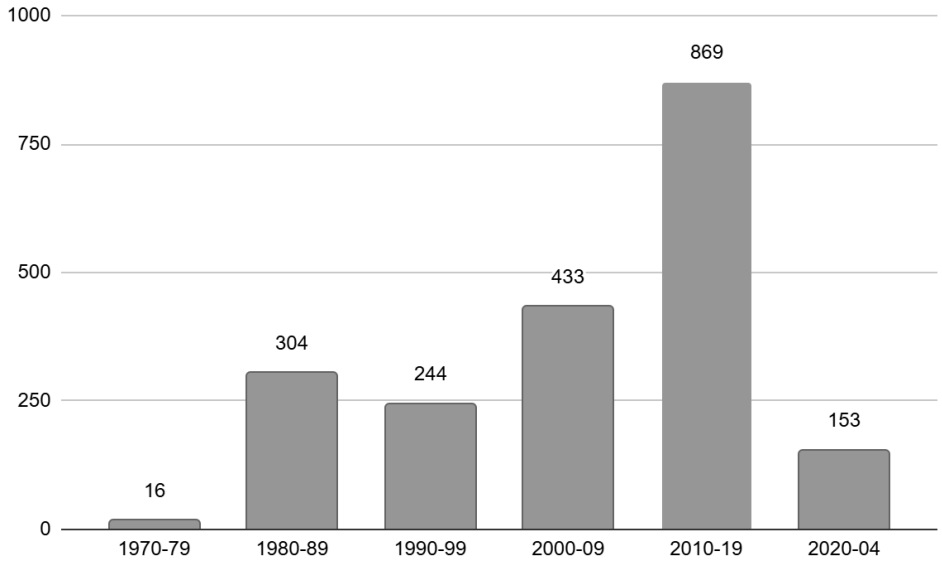
Table 2
Aggregation of effect sizes from different designs.

Mixed effect sizes	Influences	% of total
Between-, within-subject designs, and correlational research	30	10%
Between- and within-subject designs	51	17%
Between-subject design and correlational research	64	21%
Within-subject design and correlational research	7	2%
Missing information in order to assess	5	2%

4. To what extent have meta-analyses with a wide publication year range been mixed?

Figure 1 describes the distribution publication year of included meta-analyses. As we can see that around 50% of the included meta-analyses are published after the year 2010. At first glance this may look reasonable, however, when examining the distribution in each influence, it is not plausible that it is representative of the most up-to-date evidence. Indeed, during our assessment we discovered that *Visible Learning* includes old meta-analyses even when newer, updated ones exist.

Figure 1
Descriptives of publication year including meta-analyses from all 322 influences



As mentioned in the introduction, some interventions might not be affected by the time they were conducted, for example well researched constructs such as phonics. But other topics should definitely be updated with the latest studies. An example of where the year of publication can be affecting the results are in influences that focus on technology in school. Examples of influences containing technology or computers can be found in table 3. In some of the examples in table 3 the majority of the studies are conducted in the 1980s.

Table 3
Influences including technology and publication year of included meta-analyses

Influence	1980	1990	2000	2010	2020
Technology in mathematics	5	4	4	6	6
Technology in other subjects	-	1	2	-	-
Technology in reading / literacy	-	-	10	6	1
Technology in science	3	-	2	1	-
Technology in small groups	-	-	3	-	-
Technology in writing	1	1	1	1	-
Technology with college students	3	1	3	8	-
Technology with elementary students	3	3	-	-	-
Technology with high school students	6	2	1	-	-
Technology with learning needs students	2	1	2	-	-
Television hours	2	-	-	1	-

5. What publication types (conference papers, unpublished doctoral theses, peer-reviewed articles) are included in Visible Learning?

As expected in any review, the most common publication type is peer-reviewed articles, followed by unpublished theses or dissertations. When only describing *Visible Learning* in summary descriptives, these numbers do not appear that alarming as the addition of grey literature can combat publication bias. Table 4 presents the descriptives for all publication types included in *Visible Learning.*

Table 4
Descriptives of publication type from all 322 influences

Publication type	n	% of total
Journal article	1520	76%
Unpublished theses	275	14%
Conference papers	97	5%
Reports	42	2%
Books / Chapters	44	2%
Others	11	1%

However, when we look closer to some influences we can see that 24 influences only aggregate non-peer reviewed material. This includes well-known influences, such as 'Middle school interventions' which was based on a single unpublished dissertation from 2011 and 'School size' which only included one unpublished dissertation from 1991.

6. To what extent can the reported statistics in Visible Learning (e.g. effect sizes, numbers of studies, or participants) be reproduced?

As mentioned in the Methods, we tried to reproduce each influence until failure. We were only able to reproduce two influences: 'Student feeling disliked' and 'Within-class grouping'. The first influence included one meta-analysis and we were able to extract the same effect size that matched PICO, the number of participants, studies, and effects. The influence 'Within-class grouping' included three meta-analyses but we were unable to retrieve two papers, hence this counts as a full reproduction of the influence. In total, we were able to reproduce the first randomized meta-analysis of 12 other influences but we failed to reproduce the second meta-analysis in the order.

In total, 312 meta-analyses were attempted to be reproduced from the 301 influences. 44% (n = 133) of the reported effect sizes could not be found in the original meta-analysis, 33% (n = 99) of the effect sizes could be numerically reproduced, however, the effect size did not match the PICOS provided by the influence. 26% (n = 80) of the effect sizes could be found and reproduced accurately. As mentioned we were unable to find any included meta-analyses for 21 influences and they are therefore treated as missing.

In the cases where we were able to reproduce the effect size (n = 80), other extraction of statistics instead failed. The most common error was that the extracted number of participants (72%) were missing or wrongly reported, followed by the wrong number of included studies (43%) and lastly the number of effects were not correct (35%).

To summarize, *Visible Learning* is not reproducible. During our assessment, it was common for us to easily identify the correct statistics to extract, but *Visible Learning* had not done so. In other words, the reproducibility is due to poor quality in the extraction of data from the meta-analyses.

7. What is the perceived relevance of the different influences in Visible Learning, based on a panel of expert education researchers' ratings?

According to the panel, 72 influences were rated with a score of four or more out of five regarding relevance to schools. The top ten rated influences are shown in table 5. All researchers in the expert panel scored five on the influence: *'teacher estimates of achievement'*, and thus makes it the most perceived relevant influence on the list.

Table 5
Top 10 influences rated highest on relevance

Publication type	Mean	SD
Teacher estimates of achievement	5.00	0.00
Bullying	4.75	0.50
Classroom management	4.75	0.50
Vocabulary programs	4.75	0.50
Feedback	4.75	0.50
Reading comprehension programs	4.50	0.58
Principals / School leaders	4.50	0.58
Prior ability & achievement	4.50	1.00
Outlining and summarizing	4.50	0.58
Peer tutoring on tutor	4.50	0.58

The five least relevant influences according to the panel were 'Breastfeeding (M = 1, SD = 0), 'Visual perception programs' (M = 1.25, SD = 0.5), 'Full- vs. pre-term / low birth weight' (M = 1.25, SD = 0.5), 'Matching teaching to style of learning' (M = 1.25, SD = 0.5), and 'Chess instructions' (M = 1.5, SD = 0.58). In total, 35 influences were assessed with a relevance rating of two or less. These results suggest that *Visible Learning* includes both highly relevant and less relevant influences within its presented list. The highly relevant influences, however, offer a valuable basis for guiding future directions in educational research.

Exploratory findings

During the assessment of *Visible Learning*, we made a few discoveries about things that we think are worth noting. First of all, we found the influence definitions very unclear and often appearing more like topics than definitions of something that influences student achievement. Second, we found various errors not covered by our assessment. For example, we did not think to check whether all studies were actually meta-analyses, but we did notice original studies being included, as well as commentaries and other types of studies that had no place being included. As this was outside of our pre-registered assessment, and thus prone to bias, we did not systematically document these issues.

Discussion

Prior to the present research, the methodological criticism against *Visible Learning* was scattered and unsystematic. Researchers highly skilled in methods and statistics (e.g., Bergeron, 2017; Simpson, 2017) have uncovered several flaws in *Visible Learning*, but have not had the time or the resources to systematically and exhaustively assess the whole mega-study. Having spent almost four years, as well as considerable resources, we are now prepared to determine whether the previous methodological criticism was cherry-picked or just the tip of the iceberg.

Our findings are clear: *Visible Learning* is flawed to the point that the presented effect sizes are meaningless numbers. Further, the problem does not appear to be isolated to certain influences and thus easy to address, but rather systemic to the whole methodological approach of *Visible Learning*. Although we still think that the core idea of *Visible Learning* has great potential, the execution is flawed to the point that it would be less work to start over from scratch than to try and repair it.

The most severe flaws in Visible Learning

Rather than reviewing the issues in the order of assessment (following our pre-registration), we will now review them in order of how problematic they turned out to be, starting with the worst: reproducibility. We were only able to reproduce 2 out of the 301 influences, or less than 1%. This implies major problems in the data extraction and coding of the literature, making the numbers reported in *Visible Learning* unreliable to the point that they cannot be trusted.

The second most severe problem is the inclusion of irrelevant meta-analyses when synthesizing the influences. Only 19% of the 301 influences passed all our criteria by including only meta-analyses from the relevant Populations, Interventions, Comparisons and Outcomes (PICO). This means that the definitions of the influences do not accurately describe the included meta-analyses; an influence might be based on studies with an entirely different population, intervention, or even outcome, than the definition suggests. This issue also highlights substantial disregard of well-established methodological frameworks, such as the Methodological Expectations of Campbell Collaboration Intervention Reviews (MECCIR).

It was further a very common problem that influences included a substantial part of meta-analyses that were old and outdated. Whether this is a severe problem depends on the subject, but in many cases *Visible Learning* does not accurately depict the current evidence in the literature. The method of including all old meta-analyses means that *Visible Learning* is exceptionally conservative: in areas that start with meta-analyses published in the 1980s, if old primary studies are counted several times (because they appear in multiple included meta-analyses), it becomes very hard for findings of new studies to update the effect sizes in *Visible Learning*. In other words, the inclusion of old meta-analyses allows older studies to disproportionately outweigh the newer studies. This becomes a severe problem for all influences that are time-dependent or have rapid progress, such as those that involve technology. In these cases, *Visible Learning* is, essentially, showing the effect technology had several decades ago.

Less flawed areas

Close to half (49%) passed the criteria of not mixing different types of effect sizes in an unprincipled manner (i.e., combining correlational, within and between effect sizes without corrections). The results of our assessment of the mixing of study designs are slightly less alarming, with 73% passing our criteria. These figures, however, are quite far away from any conventionally acceptable error rates.

While *Visible Learning* generally did a good job at including peer reviewed material together with grey literature, this is not evenly mixed across influences. In fact, some influences only included doctoral dissertations and conference papers, despite the fact that anyone could easily verify the existence of peer-reviewed articles.

A relevant list of influences

Now, let us turn to the one area where *Visible Learning* actually passed our assessment: relevance. Only 11% of the influences were rated as irrelevant, and 23% were rated as highly relevant. This suggests that *Visible Learning* can function as a meta-scoping review that provides a comprehensive overview of the potential influences of student achievement. Yet, as a general observation, during the quality assessment, we often found the definitions of the influences to be quite unclear and reading more like loose topics. Alas, a systematic assessment of the definitions' wording was beyond the scope of this research.

Implications for researchers

The very poor methodological quality and almost non-existent reproducibility of *Visible Learning* means that researchers cannot rely on the reported effect sizes as a basis to conduct future research, or cite this as a source of evidence for an influence on student achievement. The meta-analyses for each influence are so poorly collected and combined that a researcher is better off simply searching a reputable database (e.g., ERIC) for the influence and reading the most recent systematic reviews.

Visible Learning does, however, succeed in being quite an exhaustive list of relevant influences on student achievement. It can be useful for researchers to consult this when planning studies, as long as they ignore the statistics that are presented. Further, by consulting our shiny app or appendix, they can get a better idea of what areas to pursue if they aim to conduct new systematic reviews or meta-reviews.

Implications for school professionals and policymakers

Very few, if any, of the influences listed in *Visible Learning* are of high enough methodological quality to inform educational policy, or be useful for professionals in school. Close to none of the effect sizes are reproducible, and even if they were, it would not matter as they typically include widely different studies that cannot be meaningfully combined. Our message to practitioners is clear: do not rely on *Visible Learning* for anything other than as a list of potential influences of student achievement. The reported effect sizes are, at best, unreliable and incorrect; at worst, outright misleading.

Limitations

This study provides a comprehensive assessment of *Visible Learning*, yet some limitations must be acknowledged. First, the heterogeneity of the assessed material was a challenge in terms of interrater reliability. Although consensus was reached through discussion, the subjective aspects of the assessment may have influenced the results. Yet, to a large extent, this is due to the vaguely defined influences that makes the assessment of PICOS adherence difficult.

Second, the inability to access all referenced meta-analyses due to them being incomplete or unavailable, limits the comprehensiveness of our quality assessment. However, we believe that this gap in data retrieval does not affect the findings significantly, given the severity of the findings. If anything, finding and including these meta-analyses would likely make our findings worse, as the quality of the missing studies can be expected to be lower than the ones included.

Third, we discovered during the process that there likely had been no systematic searches for meta-analyses in *Visible Learning*. In hindsight, this seems like an obvious thing to include in our assessment. That said, we are unsure how feasible it would have been to assess the searches, as it would have taken many hours per influence to conduct proper systematic searches to compare to.

Future research

When we began this project in 2021, we were quite hopeful that an assessment of *Visible Learning* would, at least, discover a core of highly quality influences that were also highly relevant, and that it would be feasible to improve the influences that lacked in quality but were relevant. Today, we are skeptical about whether this is a good way forward. The flaws in *Visible Learning* are systemic and span the entire research process. Although we still believe in the overall idea to gather and synthesize the evidence for all influences on student achievement, we

do not think that continuing the work with a basis in *Visible Learning* is an efficient way forward.

This raises the question of what should be done instead. Indeed, *Visible Learning* did fill an important function when released. Today, other sources have, to some extent, caught up. The scientific journal *Campbell Systematic Reviews* are following the highest methodological standards, and are thus an excellent source for researchers to consult, and also to contribute too. These systematic reviews are also useful for school professionals and policymakers, as they include plain language summaries.

Conclusion

The introduction of Hattie's *Visible Learning* was nothing short of revolutionary. Researchers and school professionals could, at last, get a definitive overview of what influences student achievement. We applaud this bold idea, and think it deserves credit as one of the major ideas in educational research. Yet, as with many innovations, it is not without early-stage flaws.

As promising as the idea of *Visible Learning* is, the methodological quality and reproducibility is entirely unacceptable for anything that informs educational practices. It has now been fifteen years since John Hattie launched his first mega-study. We hope that the research community is finally ready to accept the overall idea as useful, and step up to the challenge to deliver an easy-to-comprehend overview on how to improve student achievement, while also maintaining the highest levels of methodological quality, reproducibility, and relevance.

References

Aloe, A. M., Dewidar, O., Hennessy, E. A., Pigott, T., Stewart, G., Welch, V., Wilson, D. B., & Campbell MECCIR Working Group. (2024). Campbell Standards: Modernizing Campbell's Methodologic Expectations for Campbell Collaboration Intervention Reviews (MECCIR). *Campbell Systematic Reviews, 20*(4), e1445. https://doi.org/10.1002/cl2.1445

Allison, D. B., Brown, A. W., George, B. J., & Kaiser, K. A. (2016). Reproducibility: A tragedy of errors. *Nature, 530*, 27–29.

Arnold, I. (2011). John Hattie: Visible learning: A synthesis of over 800 meta-analyses relating to achievement. *International Review of Education, 57*(1), 219. https://doi.org/10.1007/s11159-011-9198-8

Berger, M. A. (1982, April 17–20). Class size is not the issue [Paper presentation]. National School Boards Association.

Bergeron, P.-J. (2017). How to engage in pseudoscience with real data: A criticism of John Hattie's arguments in "Visible Learning" from the perspective of a statistician. *McGill Journal of Education, 52*(1), 237–246.

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (Eds.). (2013). *Introduction to meta-analysis*. Wiley.

Burns, M. K., & Symington, T. (2002). A Meta-analysis of Prereferral Intervention Teams: Student and Systemic Outcomes. *Journal of School Psychology, 40*(5), 437–447. https://doi.org/10.1016/S0022-4405(02)00106-1

Campbell Collaboration. (2014). Campbell collaboration systematic reviews: Policies and guidelines (No. 1.4). The Campbell Collaboration. https://doi.org/10.4073/cpg.2016.1

Carlsson, R., Batinović, L., Hyltse, N., Kalmendal, A., Nordström, T., & Topor, M. (2024). A Beginner's Guide to Open and Reproducible Systematic Reviews in Psychology. *Collabra: Psychology, 10*(1), 126218. https://doi.org/10.1525/collabra.126218

Cheung, A. C. K., & Slavin, R. E. (2016). How methodological features affect effect sizes in education. *Educational Researcher, 45*(5), 283–292. https://doi.org/10.3102/0013189X16656615

Cicchetti, D. V. (1994). Guidelines, criteria, and rules of thumb for evaluating normed and standardized assessment instruments in psychology. *Psychological Assessment, 6*(4), 284–290. https://doi.org/10.1037/1040-3590.6.4.284

Corwin Visible Learning Plus. (2023). *Global Research Database: Visible Learning Meta X*. https://www.visiblelearningmetax.com/influences

Derry, C., Derry, S., McQuay, H., & Moore, R. (2006). Systematic review of systematic reviews of acupuncture published 1996–2005. *Clinical Medicine, 6*(4), 381–386. https://doi.org/10.7861/clinmedicine.6-4-381

Dewidar, O., Hennessy, E. A., Pigott, T., Stewart, G., Welch, V., & Wilson, D. B. (2023). Methodological expectations of Campbell Collaboration Intervention Reviews (MECCIR): 2023 Update. *Campbell Collaboration*. https://doi.org/10.17605/OSF.IO/KCSPX

DeWitt, P. (2018). Using the Visible Learning Research to Influence Collaborative Leadership. *Education Sciences, 8*(4), 219. https://doi.org/10.3390/educsci8040219

Dowle, M., & Srinivasan, A. (2022). *data.table: Extension of data.frame* (Version 1.14.6). https://CRAN.R-project.org/package=data.table

Firke, S. (2023). *janitor: Simple tools for examining and cleaning dirty data* (Version 2.2.0). https://CRAN.R-project.org/package=janitor

Gøtzsche, P. C., Hróbjartsson, A., Marić, K., & Tendal, B. (2007). Data extraction errors in meta-analyses that use standardized mean differences. *JAMA, 298*(4). https://doi.org/10.1001/jama.298.4.430

Graham, S., Liu, X., Bartlett, B., Ng, C., Harris, K. R., Aitken, A., Barkel, A., Kavanaugh, C., & Talukdar, J. (2018). Reading for Writing: A Meta-Analysis of the Impact of Reading Interventions on Writing. *Review of Educational Research, 88*(2), 243–284. https://doi.org/10.3102/0034654317746927

Hansen, O., Lekhal, R., Nordahl, T., Norlund, A., Persson, B., & Qvortrup, L. (Eds.). (2015). *Temanummer om synlig læring*. [Special Issue] *Paideia, 9*.

Hattie, J. (2009). *Visible learning: A synthesis of over 800 meta-analyses relating to achievement*. Routledge.

Hattie, J. (2023). *Visible Learning: The Sequel: A Synthesis of Over 2,100 Meta-Analyses Relating to Achievement* (1st ed.). Routledge. https://doi.org/10.4324/9781003380542

Higgins J. P. T., Thomas, J., Chandler, J., Cumpston, M., Li, T., Page, M. J., & Welch, V. A. (Eds.). (2024). *Cochrane Handbook for Systematic Reviews of Interventions version 6.5* (updated August 2024). Cochrane. Available from www.training.cochrane.org/handbook

Huang, C. Y., Costeines, J., Kaufman, J. S., & Ayala, C. (2014). Parenting Stress, Social Support, and Depression for Ethnic Minority Adolescent Mothers: Impact on Child Development. *Journal of Child and Family Studies, 23*(2), 255–262. https://doi.org/10.1007/s10826-013-9807-1

Knudsen, H. (2017). John Hattie: I'm a statistician, I'm not a theoretician. *Nordic Journal of Studies in Educational Policy, 3*(3), 253–261. https://doi.org/10.1080/20020317.2017.1415048

Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine, 15*(2), 155–163. https://doi.org/10.1016/j.jcm.2016.02.012

Lakens, D., Hilgard, J., & Staaks, J. (2016). On the reproducibility of meta-analyses: Six practical recommendations. *BMC Psychology, 4*(1), 24. https://doi.org/10.1186/s40359-016-0126-3

McKnight, L., & Whitburn, B. (2020). Seven reasons to question the hegemony of Visible Learning. *Discourse: Studies in the Cultural Politics of Education, 41*(1), 32–44. https://doi.org/10.1080/01596306.2018.1480474

McShane, B. B., Böckenholt, U., & Hansen, K. T. (2016). Adjusting for publication bias in meta-analysis: An evaluation of selection methods and some cautionary notes. *Perspectives on Psychological Science, 11*(5), 730–749. https://doi.org/10.1177/1745691616662243

Morris, S. B., & DeShon, R. P. (2002). Combining effect size estimates in meta-analysis with repeated measures and independent-groups designs. *Psychological Methods, 7*(1), 105–125. https://doi.org/10.1037/1082-989X.7.1.105

Nejman, F. (2023). John Hatties metoder – en pedagogisk fluga. *Ämnesläraren – Svenska, språk mm*. https://www.vilarare.se/amneslararen-svenska-sprak/ny-kategori/john-hatties-metoder--en-pedagogisk-fluga/

Nielsen, K., & Klitmøller, J. (2021). Blind spots in visible learning: A critique of John Hattie as an educational theorist. *Nordic Psychology, 73*(3), 268–283. https://doi.org/10.1080/19012276.2021.1962731

Nilvius, C., Carlsson, R., Fälth, L., & Nordström, T. (2021). Tier 2 interventions within the RtI-model for developing students' word decoding – A systematic review and meta-analysis. *Cogent Education, 8*(1), 1994105. https://doi.org/10.1080/2331186X.2021.1994105

Ooms, J. (2023). *pdftools: Text Extraction, Rendering and Converting of PDF Documents*. R package version 3.4.0, https://CRAN.R-project.org/package=pdftools.

Revelle, W. (2024). *psych: Procedures for psychological, psychometric, and personality research* (Version 2.4.3). Northwestern University, Evanston, Illinois. https://CRAN.R-project.org/package=psych

Rømer, T. A. (2019). A critique of John Hattie's theory of Visible Learning. *Educational Philosophy and Theory, 51*(6), 587–598. https://doi.org/10.1080/00131857.2018.1488216

R Core Team. (2021). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.r-project.org/

Simonsohn, U., Nelson, L. D., & Simmons, J. P. (2014). p-Curve and Effect Size: Correcting for Publication Bias Using Only Significant Results. *Perspectives on Psychological Science, 9*(6), 666–681. https://doi.org/10.1177/1745691614553988

Simpson, A. (2017). The misdirection of public policy: Comparing and combining standardised effect sizes. *Journal of Education Policy, 32*(4), 450–466. https://doi.org/10.1080/02680939.2017.1280183

Slavin, R. (2018, June 21). John Hattie is Wrong. *Robert Slavin's Blog*. https://robertslavinsblog.wordpress.com/2018/06/21/john-hattie-is-wrong/

Snook, I., O'Neill, J., Clark, J., & Openshaw, R. (2009). Invisible learnings? A commentary on John Hattie's book: *Visible Learning: A synthesis of over 800 meta-analyses relating to achievement*. *New Zealand Journal of Educational Studies Preview Publication Details, 44*(1), 93–106.

Stuebing, K. K., Barth, A. E., Trahan, L. H., Reddy, R. R., Miciak, J., & Fletcher, J. M. (2015). Are Child Cognitive Characteristics Strong Predictors of Responses to Intervention? A Meta-Analysis. *Review of Educational Research, 85*(3), 395–429. https://doi.org/10.3102/0034654314555996

Tran, L., Sanchez, T., Arellano, B., & Lee Swanson, H. (2011). A Meta-Analysis of the RTI Literature for Children at Risk for Reading Disabilities. *Journal of Learning Disabilities, 44*(3), 283–295. https://doi.org/10.1177/0022219410378447

Vaidyanathan, R., Russell, K., & Watts, G. (2016). *sparkline: 'jQuery' sparkline 'htmlwidget'* (Version 2.0). https://CRAN.R-project.org/package=sparkline

Vaidyanathan, R., Xie, Y., Allaire, J., Cheng, J., Sievert, C., & Russell, K. (2023). *htmlwidgets: HTML widgets for R* (Version 1.6.4). https://CRAN.R-project.org/package=htmlwidgets

Wickham, H. (2022). *stringr: Simple, consistent wrappers for common string operations* (Version 1.5.0). https://CRAN.R-project.org/package=stringr

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., & Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software, 4*(43), 1686. https://doi.org/10.21105/joss.01686

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). *dplyr: A grammar of data manipulation* (Version 1.1.3). https://CRAN.R-project.org/package=dplyr

Wrigley, T. (2018). The power of 'evidence': Reliable science or a set of blunt tools? *British Educational Research Journal, 44*(3), 359–376. https://doi.org/10.1002/berj.3338